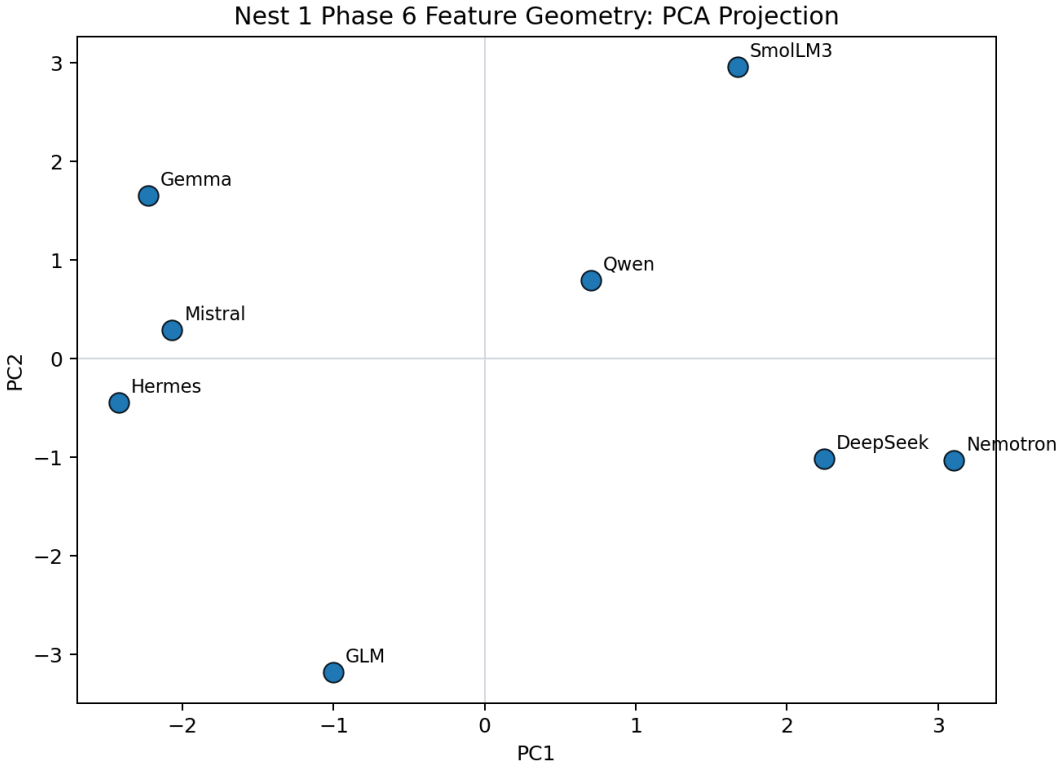
Nest 1 Real-Trace Foundation Evidence Pack

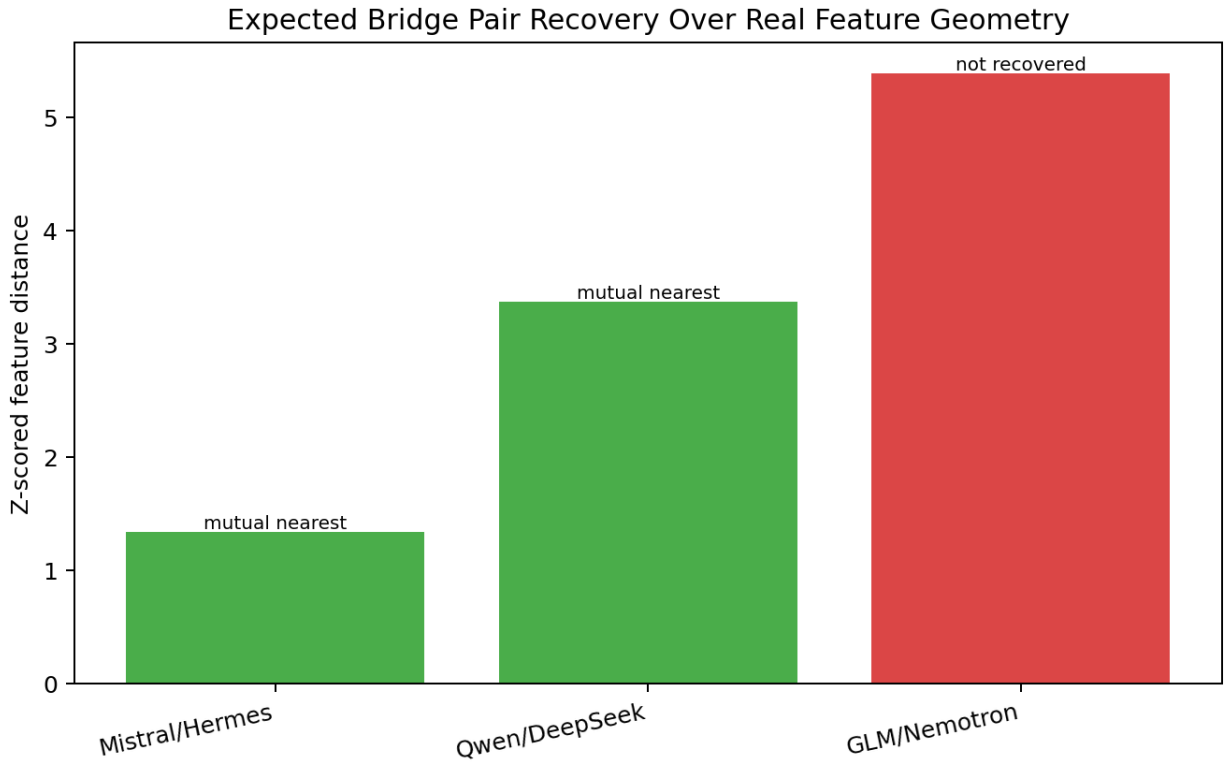
Generated from real exported AI and hardware artifacts. No toy rows are used in this pack.

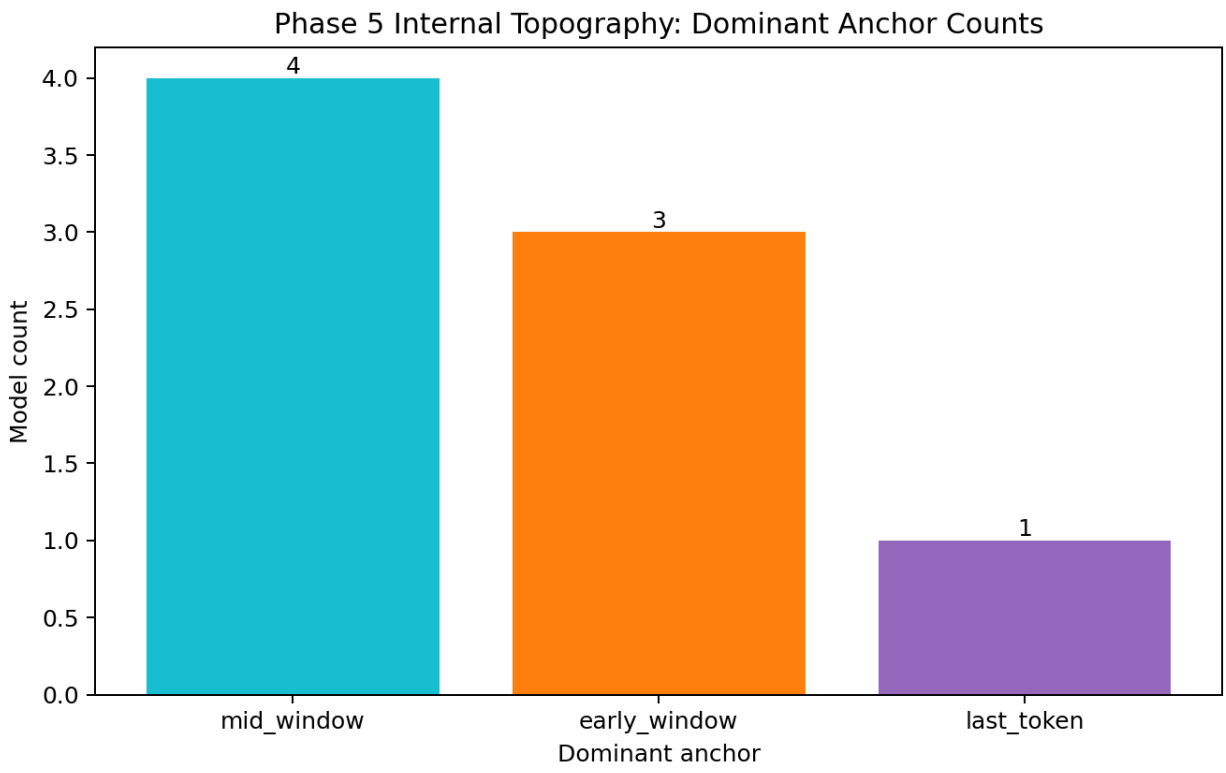
Branch	Status	Key read
linear_algebra_geometry	real_trace_run	Phase 6 normalized feature matrix supports PCA/SVD, pairwise distance, cosine, and bridge-pair checks.
tensor_information	real_trace_run	Model x feature matrix gives a real tensor/information summary, but raw activation tensors would strengther
statistics_probability	real_trace_run	Phase 2 rerun matrix remains the strongest real STAT/PROB surface.
graph_topology_lite	real_trace_run_limited	KNN graph over real feature vectors is a topology-lite pass; stronger topology needs raw hidden-state point
topography_bridge	real_trace_run	Phase 5 context-to-readout anchors give a real topographic bridge over internal model traces.
dynamical_systems	real_trace_run	Layerwise target/control delta trajectories provide a real dynamics-over-depth pass.
numerical_group_symmetry	real_hardware_run	Phase 9D hardware repeatability gives a real numerical/backend and sign-symmetry surface.
optimization	blocked_new_benchmark_needed	Needs a declared mirror-guided optimization benchmark against naive/random/standard baselines.
control_theory	blocked_trace_needed	Needs exported real LSPS/Oracle transition traces.
category_composition	design_lane	Needs a measured transfer test showing structure preserved between two real nests.

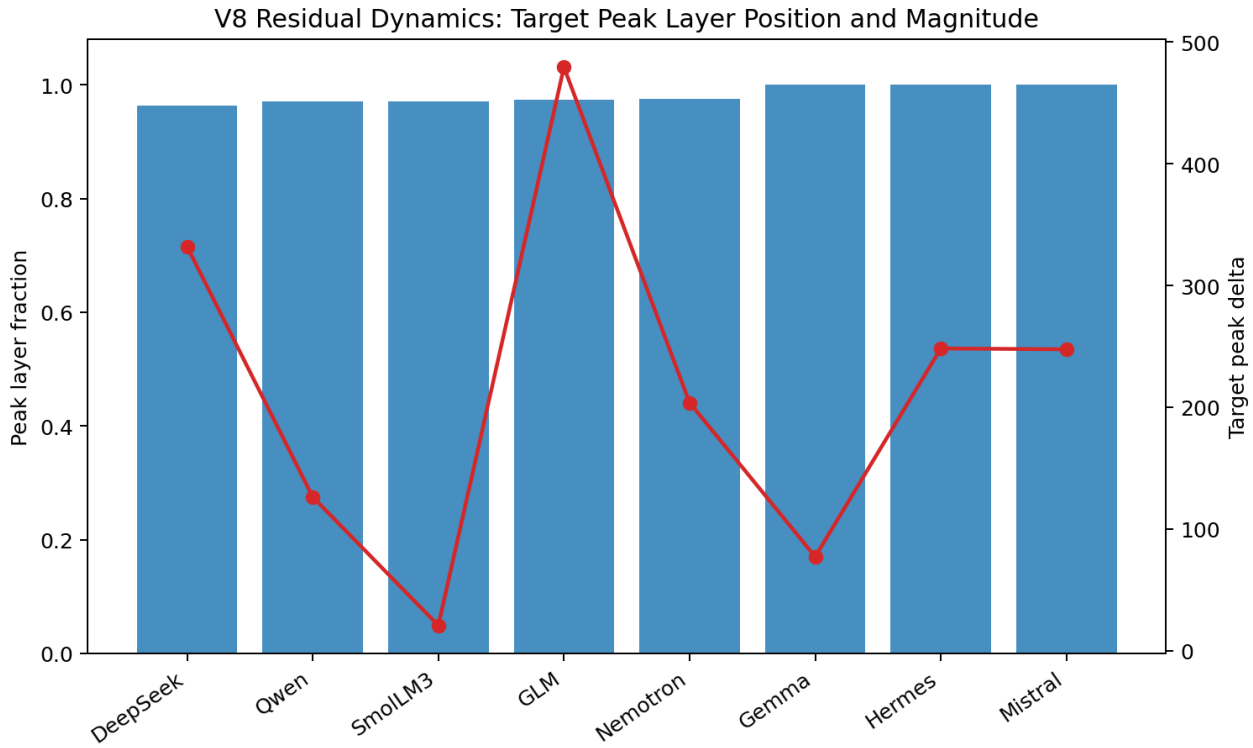
Clean read: Nest 1 now has a real foundation pass over the existing AI/hardware stack. Linear algebra/geometry, tensor/information, statistics/probability, topography, layer dynamics, and hardware sign stability are grounded in real artifacts. Optimization, control, deeper topology, and composition remain explicit next-runner lanes.

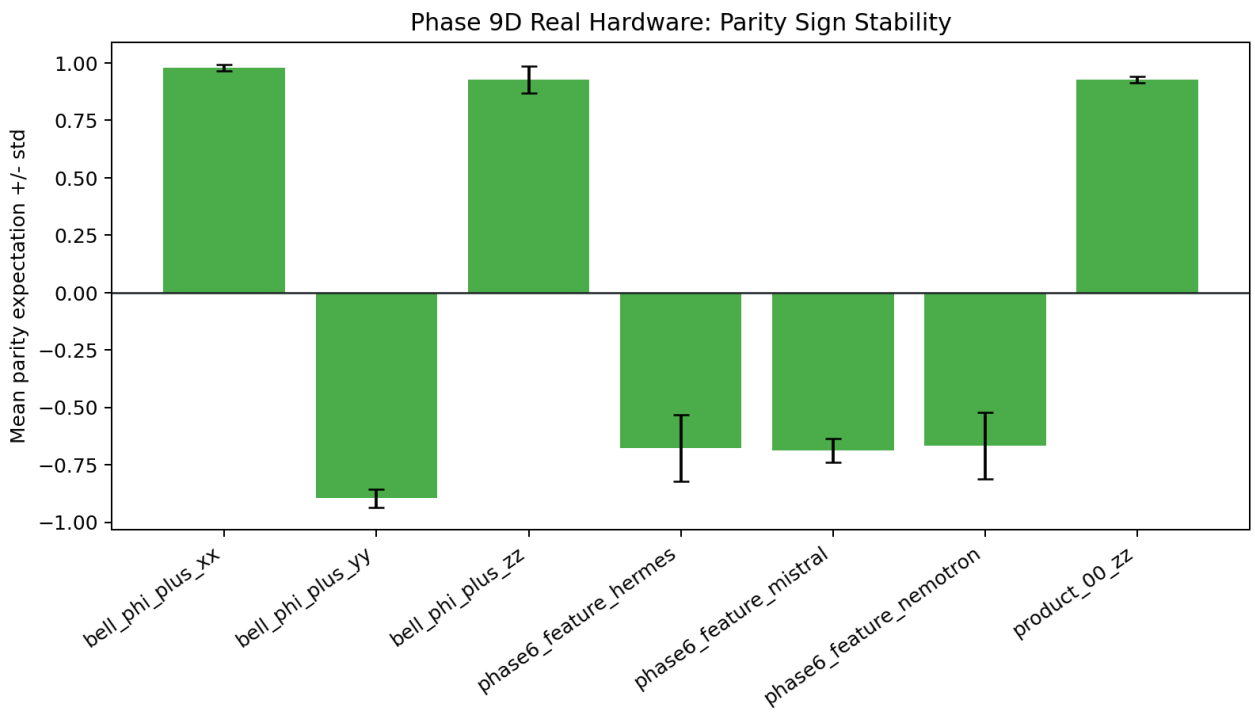
This uses real exported V8 and hardware artifacts. Some lanes are limited by the currently exported scalar summaries and still need raw hidden-state point clouds, LSPS traces, or new benchmarks before stronger claims.




Expected Bridge Pair Recovery Over Real Feature Geometry
Z-scored feature distance
0
1
2
3
4
5
mutual nearest
mutual nearest
not recovered
Mistral/Hermes
Qwen/DeepSeek
GLM/Nemotron


Phase 5 Internal Topography: Dominant Anchor Counts
Model count
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4
3
1
mid_window
early_window
last_token
Dominant anchor


V8 Residual Dynamics: Target Peak Layer Position and Magnitude
Peak layer fraction
Target peak delta
0.0
0.2
0.4
0.6
0.8
1.0
0
100
200
300
400
500
DeepSeek
Qwen
SmolLM3
GLM
Nemotron
Gemma
Hermes
Mistral


Phase 9D Real Hardware: Parity Sign Stability
Mean parity expectation +/- std
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
bell_phi_plus_xx
bell_phi_plus_yy
bell_phi_plus_zz
phase6_feature_hermes
phase6_feature_mistral
phase6_feature_nemotron
product_00_zz